FLEET TEAM DIGEST

737MAX-FTD-32-17017

Issue Title : Corrosion on Messier-Bugatti-Dowty Main Gear Wheels

Airplane Model	ATA	Minor Model(s)
737MAX	3245-11	-7, -8
Other Model(s)	**ECCN**	
737NG	9E991	

Originated Date	Last Revised Date	Created On
05/11/2017	05/11/2017	04/25/2017
Estimated Completion Date	**Next Update Date**	

Applicability

All 737MAX airplanes using Messier-Bugatti-Dowty (MBD) main gear wheels.

Note: Not applicable to -9 as it does not utilize subject wheel.

Description

A few 737NG operators have reported corrosion on MBD main gear wheels. The corrosion is typically found in these locations:

- Tie bolt hole inside diameter

- The wheel half mating faces, near the tie bolt holes

- Tie bolt hole radii on each wheel half (mating face side)

The attached photos show several examples.

The 737 MBD wheels have been in service since 2008. However the corrosion problems became more widely reported in 2013 and in early 2014 than the first 4 years of service.

Hours/Cycles

The number of cycles on the wheel halves when the corrosion was found has varied widely. However, recently some of the affected wheel halves were nearly new (only a few tire changes) when the corrosion was discovered.

Background

MBD and Boeing have performed a comprehensive study to determine the root cause(s) of the corrosion.

Based on lab testing and other investigation results, MBD and Boeing have made the following observations / conclusions:

1) The protective anodize layer is too thin in and around the bolt hole areas of the wheel.

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2) The graphite-based thread lubricants specified in the CMM for the tie bolts can promote corrosion in and around the bolt holes if the paint and protective coatings are not maintained.

3) There is a very small motion that occurs between the wheel halves as the wheel rolls under load. This small motion abrades the paint and chemical conversion coatings on the mating faces, leaving the bare aluminum prone to corrosion attack and fretting damage. As a result, the paint and chemical conversion coating on the mating faces need to be maintained at EACH tire change.

Status

Closed.

Interim Action

See "Operator Action" section below.

Final Action

MBD made changes to their anodize process to improve the thickness and consistency of the anodize layer. All wheels manufactured after late December 2014 will have the improved anodize. The first wheel serial number with the improved anodize process will be "A001C20626" and on. Note that the serial number convention will be changed coincident with the anodize change (e.g. A001 instead of fully numeric).

MBD developed more generous repairs for corrosion in and around the bolt holes.

MBD introduced an improved paint system that has better adhesion and better corrosion protection than the earlier paint scheme. The new paint has been applied to all wheel halves manufactured as of late March, 2014. MBD has released service bulletins allowing this new paint to be used on the 737 main wheels, nose wheels, and brake piston housings.

MBD evaluated a silicone based MIL-G-25013 lubricant as an alternative to the graphite-based lubricants currently used on this wheel. The evaluation was completed; Product Deviation Authorization (PDA) N0. 036-15. CMM 32-49-83 was issued, and MBD SB C20626-32-007 provides approval for the new lubricant as of 1 March 2016.

Operator Action

Boeing provides the following suggestions to help prevent the corrosion:

- Operators should thoroughly clean off all remaining traces of primer on the mating faces and tie bolt hole areas of the wheels at EACH tire change. Mild abrasives (such as Scotch-Brite pads or similar) have been effective at cleaning off these surfaces. One operator has had success with Scotch-Brite Roloc 50 plastic bristle brushes in a pneumatic rotary tool.

- Remove corrosion per CMM 32-49-83. MBD has issued several Product Deviation Authorizations (PDA's) to individual operators that offer corrosion repairs beyond what is currently shown in the CMM. Operators should contact MBD for repair solutions that may allow operators to re-use damaged wheel halves.

- If operators are experiencing corrosion in these areas, MBD recommends using Deoxidine (a mild phosphoric acid-based solution that neutralizes corrosion) and applying it to affected surfaces. Deoxidine is discussed in MBD Standard Practices Manual (SPM) 32-09-01, section 14.

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- After the corrosion removal, apply chemical conversion coatings (Alodine) per MBD CMM 32-49-83.

- Reapply primer to the mating faces with one of the approved primers. Note that replacing the primer is very important since it acts as a sacrificial barrier to protect against the 'fretting' motion between the two wheel halves. Without the paint layer, the small motion between the wheel halves can cause damage to the chemical conversion coatings and the base aluminum. Some operators have fabricated paint masking devices so the mating faces can be re-primed at tire change without getting overspray on other parts of the wheel that have topcoat.

- Operators may wish to migrate to one of the more durable paint schemes available for this wheel. The referenced MBD service bulletin provides an improved paint scheme that became the production paint in mid-2014. Also, MBD CMM 32-49-83 allows BMS 10-79 Type III primer and BMS 10-60 Type II topcoat. This is the standard paint scheme used on landing gear structural components on Boeing airplanes and has proved very durable (see CMM 32-49-83, Repair 7 – 'Protection').

- Operators may wish to consider switching to use of the new silicon based lubricant per MBD SB C20626-32-007.

Supplier Information

Messier-Bugatti-Dowty

Wheels and Brakes

Zone Aeronautique Louis Brequet

78140 Velizy-Villcoublay

France

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Service Bulletin (SB)	C20626-32-008 (Messier-Bugatti-Dowty)		
Component Maintenance Manual (CMM)	MBD 32-49-83		
Other	Standard Practices Manual (MBD) 32-09-01		
Service Bulletin (SB)	C20626-32-007 (Messier-Bugatti-Dowty)		03/01/2016

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Wheel, Main Landing Gear	S277A016-551	Messier-Bugatti-Dowty	C20626200

Related Categories

- Component Issues : Issues that occur during component maintenance or have an impact on component reliability

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Attachments:

a) Corrosion Example 3

b) Corrosion Example 2

c) Corrosion Example 1

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